EMG Baseline Reproduction — Stage A (MAP parity) + Stage B (BSC project baseline)

Anchor paper: Rahimi, Benatti, Kanerva, Benini, Rabaey, "*Hyperdimensional Biosignal Processing: A Case Study for EMG-based Hand Gesture Recognition,*" IEEE ICRC 2016. Authors' code + data: `github.com/abbas-rahimi/HDC-EMG` (GPLv3).

Data: authors' bundled `dataset.mat` — 5 subjects, 4 EMG channels, enveloped signal quantized to 21 levels (`int(value)` , range 0–20), 5 gesture classes, ~98k–149k samples/subject. No external download required.

What was run: faithful NumPy ports of the authors' MATLAB reference. - Stage A — `repro/stage_a_icrc.py` (ports `ICRC.m` + `generatePaperFigures.m`). - Stage B — `repro/stage_b_bsc.py` (the binary model the 1024-bit RTL implements).

Both use the paper protocol: 25% training fraction, conditional-add / majority training, item memory seeded once, best n-gram per subject (S1=4, S2=4, S3=3, S4=5, S5=4), spatiotemporal downsampling 250 (S5=50).

1. Headline result

Configuration	Stage A — MAP, D=10000, cosine	Stage B — BSC, D=1024, Hamming	Paper
Spatial only (N=1)	**90.36 %**	**90.12 %**	90.8 %
Spatiotemporal (best N)	**96.04 %**	**89.77 %**	97.8 %

Takeaways 1. Stage A reproduces the published spatial number almost exactly (90.36 % vs 90.8 %). The spatiotemporal number (96.04 % vs 97.8 %) lands within ~2 pts; the residual is the random basis differing from MATLAB's Mersenne-Twister RNG (it cannot be byte-matched in NumPy) plus the window-slicing tie-breaks. 2. **The project's binary BSC model at D=1024 matches MAP/D=10000 for the spatial task (90.12 % vs 90.36 %)** — i.e. a 10x smaller, XOR-only, popcount-classified design loses essentially nothing on the spatial encoding. 3. The temporal channel is where binary pays a price: majority-bundled n-grams (BSC) reach only ~89.8 % at D=1024 vs 96 % for MAP. This MAP-vs-BSC temporal gap is a concrete, honest result to report.

2. Stage A — literal parity (MAP / bipolar / D=10000 / cosine)

Per-subject accuracy:

Subject	Spatial (N=1)	Spatiotemporal
S1	96.49 %	100.00 % (N=4)
S2	88.13 %	94.94 % (N=4)

Subject	Spatial (N=1)	Spatiotemporal
S3	94.37 %	97.47 % (N=3)
S4	85.44 %	95.00 % (N=5)
S5	87.37 %	92.77 % (N=4)
Mean	**90.36 %**	**96.04 %**

Model: bipolar `{-1,+1}`; bind = element-wise multiply; bundle = integer add; permute = cyclic shift by 1; similarity = cosine; conditional-add training (`cos < 0.9`). Runtime ≈ 22 s for all 5 subjects, both modes (prediction is vectorised via the linear decomposition `dot(AM, sigHV)=Σ_c P[label,c,q_c]`).

Verdict: the "published number" checkpoint is met. This is the parity anchor.

3. Stage B — project baseline (BSC / binary / Hamming), D-sweep

Same envelope encoding, re-cast in the model the RTL implements: binary `{0,1}`, **bind = XOR**, **permute = cyclic shift**, **bundle = thresholded majority (tie -> 0)**, **similarity = Hamming/popcount**. Spatial record `R = majority_c( iM[c] XOR CiM_c(v_c) )`; n-gram `G = majority_t( rho^t R[t] )`; class prototype = majority of training n-grams; classify by minimum Hamming distance.

D	Spatial mean	Spatiotemporal mean
256	90.55 %	87.39 %
512	90.06 %	88.11 %
1024	**90.12 %**	**89.77 %**
2048	90.86 %	89.65 %
4096	90.63 %	90.82 %
10000	90.59 %	90.90 %

Per-subject at D=1024 (spatial): S1 93.7, S2 89.5, S3 90.1, S4 88.6, S5 88.6.

Observations - **Spatial accuracy is essentially flat (~90–91 %) from D=256 to D=10000.** Strong evidence that the 1024-bit datapath is more than enough for the spatial task — directly supports Hook A's claim that D can be shrunk aggressively. - **Spatiotemporal accuracy rises with D** (87.4 % -> 90.9 %): the temporal binding is the dimension-hungry part, because permute+majority is noisier than MAP's permute+multiply. This is the trade-off the Pareto study should map.

4. What this gives the project

- A reproduced, citable baseline (Stage A) and an RTL-faithful binary baseline (Stage B) on identical data — both runnable from `python_ref/repro/`.

- **Project headline baseline: D=1024 BSC spatial = 90.12 %**, on par with the bipolar D=10000 result — a strong, defensible starting point.
- A clean MAP-vs-BSC and spatial-vs-temporal gap to motivate the contributions (Hook A dimension/precision Pareto; informed vs random pruning; cross-subject masks all build on this binary model and this data).

5. Reproduce

```
cd python_ref
python repro/stage_a_icrc.py --mode both          # Stage A parity, ~22 s
python repro/stage_b_bsc.py  --mode both          # Stage B D-sweep, ~2 min
# results -> repro/stage_a_results.json, repro/stage_b_results.json
```

6. Honest caveats

- NumPy RNG != MATLAB RNG, so the random basis (and therefore exact accuracy) differs slightly run-to-run; means are stable to <1 pt across seeds.
- Stage A spatiotemporal (96.0 %) is ~1.8 pts under the paper's 97.8 %; averaging several seeds / matching the exact window margins would close most of it. The spatial parity (the simplest, least ambiguous number) is the firm checkpoint.
- Stage B uses a single majority prototype per class (binary AM), matching the RTL popcount-AM, rather than Stage A's integer accumulator.

Status: Stage A DONE (parity met). Stage B DONE (project baseline recorded, D-sweep captured for Hook A). Next: freeze config (seed, D, levels, N, train fraction) into a config file (M16/M17), then wire Stage B into the RTL co-simulation path.